

1. Administrative & Study Identification

Full Study Title: A Study of Inotuzumab Ozogamicin in Chinese Patients With Relapsed or Refractory Acute Lymphoblastic Leukemia
Short Title/Acronym: Inotuzumab Ozogamicin in Chinese R/R ALL
Protocol Number: B1931034 **NCT Number:** NCT05687032 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor Name:** Pfizer **Investigational Product:** Inotuzumab ozogamicin (Besponsa) | **ATC Code:** L01FB01 **EMA Information:** [Product Information](#) **Phase of Development:** Phase 4
Medical Monitor: Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To confirm the efficacy, safety, and pharmacokinetics (PK) of inotuzumab ozogamicin in Chinese patients with relapsed or refractory CD22-positive B-cell acute lymphoblastic leukemia (ALL). **Secondary Objectives:** To evaluate Duration of Remission (DoR), minimal residual disease (MRD) negativity, progression-free survival (PFS), overall survival (OS), and to continuously monitor the safety profile (particularly the incidence of veno-occlusive disease [VOD]/sinusoidal obstruction syndrome [SOS]).

2.2 Background & Rationale To address the critical unmet medical need for targeted therapies in adult patients with relapsed or refractory B-cell ALL in mainland China. Despite standard chemotherapy, outcomes for R/R B-cell ALL remain poor. Inotuzumab ozogamicin, a CD22-directed antibody-drug conjugate (ADC), has demonstrated significant efficacy globally; this Phase 4 study aims to validate its clinical benefit and safety profile within the Chinese patient population.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Single Group Assignment) **Blinding:** Open-label
Randomization: N/A

3.2 Planned Enrollment & Duration **Target N\$:** 44 patients **Number of Sites:** Multicenter (Mainland China) **Estimated Study Start:** 24-Feb-2023 **Estimated Primary Completion:** 07-Nov-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants, age ≥ 18 years or older at screening. 2. Relapsed or refractory CD22-positive ALL. Subjects with Philadelphia chromosome-positive (Ph+) ALL must have failed standard treatment with at least one tyrosine kinase inhibitor. 3. Patients in Salvage 1 with late relapse should be deemed poor candidates for reinduction with initial therapy. 4. Patients with lymphoblastic lymphoma and bone marrow involvement $\geq 5\%$ lymphoblasts by morphologic assessment. 5. ECOG performance status 0-2 with adequate renal and hepatic function, and a negative pregnancy test for women of childbearing potential.

4.2 Exclusion Criteria 1. Subjects with isolated extramedullary relapse or active central nervous system (CNS) leukemia. 2. Prior allogeneic hematopoietic stem cell transplant (HSCT) or other anti-CD22 immunotherapy within 4 months, or active graft versus host disease (GvHD) at study entry. 3. Evidence or history of veno-occlusive disease (VOD) or sinusoidal obstruction syndrome (SOS).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Inotuzumab ozogamicin via fractionated dosing at 1.8 mg/m² per cycle (typically 0.8 mg/m² on day 1, and 0.5 mg/m² on days 8 and 15). **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Up to 6 treatment cycles.

5.2 Concomitant & Prohibited Therapy Permitted: Standard anti-infective prophylaxis and standard of care medications for co-morbidities. **Prohibited:** Concurrent administration of other investigational anti-leukemic agents, unapproved ADCs, or bridging chemotherapy not specified in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, Bone Marrow morphologic assessment, Hepatic/Renal Labs
Baseline	Day 0	Enrolment, First Dose, Vital Signs, Baseline PK	Interim
Interim	Days 1, 8, 15	of each Cycle IV Administration, Safety Labs (Liver function for VOD/SOS), PK sampling, Efficacy Assessment	End of Study
End of Study	End of Treatment / Follow-up	Final Bone Marrow assessment, Survival Follow-up, Post-treatment HSCT tracking	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of Participants With Complete Remission (CR) or Complete Remission With Incomplete Hematological Recovery (CRi) as Per Investigator's Assessment According to a Modified Cheson Criteria. * **CR:** Disappearance of leukemia as indicated by $\leq 5\%$ marrow blasts and the absence of peripheral blood leukemic blasts, with recovery of hematopoiesis defined by absolute neutrophil count (ANC) ≥ 1000 per microliter (/mCL) and platelets $\geq 100,000$ /mCL. C1 extramedullary disease (EMD) status was required (disappearance of all measurable and non-measurable EMD with the exception of lesions for which following must be true: participants with at least 1 measurable lesion, all nodal masses > 1.5 centimeter (cm) in greatest transverse diameter (GTD) at baseline regressed to ≤ 1.5 cm in GTD and nodal masses ≥ 1 cm and ≤ 1.5 cm in GTD at baseline must have regressed to ≤ 1 cm GTD or reduced by 75% in sum of products of greatest diameters (SPD). No new lesions. Spleen and other previously enlarged organs must have regressed in size and must not be palpable. All diseases were assessed using the same technique as at baseline). * **CRi:** CR except with ANC < 1000 /mCL and/or platelets $< 100,000$ /mCL.

7.2 Secondary & Exploratory Endpoints - Duration of Remission (DoR). - Percentage of Participants With Minimal Residual Disease (MRD) Negativity Among Who Achieved CR/CRi. - Progression-free Survival (PFS). - Incidence of treatment-emergent adverse events (TEAEs), specifically VOD/SOS. - Overall Survival (OS). - Pharmacokinetic (PK) parameters of inotuzumab ozogamicin.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via laboratory tests and patient reporting. Strict vigilance for liver enzyme elevation and hepatotoxicity as early indicators of VOD/SOS. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety review committee established by Pfizer.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary efficacy measures. Safety Set for AE/SAE analysis. **Sample Size Justification:** Target $N=44$ powered adequately to confirm the CR/CRi efficacy threshold and establish the safety profile within the specific regional demographic. **Handling of Missing Data:** Missing efficacy parameters will be addressed via standardized multiple imputation; patients missing follow-up without progression/death will be censored at the last known alive date.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standardized onsite monitoring frequency with central data review focused on hepatic safety signals. **Audit**

Trail: Robust tracking of eCRF locking, query resolutions, and VOD/SOS diagnostic criteria compliance.

11. Study Identifiers & Governance

Primary ID: B1931034 **Registry Number:** NCT05687032 **Sponsor/Lead**

Organization: Pfizer **Study Title:** A Study of Inotuzumab

Ozogamicin in Chinese Patients With Relapsed or Refractory Acute Lymphoblastic Leukemia **Official Regulatory Title:** A PHASE 4,

OPEN-LABEL, SINGLE-ARM, MULTICENTER STUDY OF INOTUZUMAB

OZOGAMICIN IN CHINESE ADULT PATIENTS WITH RELAPSED OR REFRACTORY CD22-POSITIVE ACUTE LYMPHOBLASTIC LEUKEMIA (ALL)

12. Clinical Framework (Study Specific)

Primary Condition: Acute Lymphoblastic Leukemia (Relapsed Acute

Lymphoblastic Leukemia) **Ontology Classifications:** MeSH: D007936 |

SNOMED: 128822004 | HPO: HP:0006721 **Diagnosis Threshold:** CD22-

positive B-cell ALL with $\geq 5\%$ lymphoblasts in bone marrow

Medical Methodology: Targeted Antibody-Drug Conjugate (ADC)

Therapy **Optimization Target:** Complete Remission (CR) and Minimal Residual Disease (MRD) negativity prior to potential HSCT

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for R/R ALL

(induction/salvage therapy bridging to potential HSCT) **Education**

Protocol (HCP): Stringent training on VOD/SOS risk factors, early detection, and liver function monitoring guidelines **Education**

Protocol (Patient): Onsite education regarding signs of hepatic toxicity, infection prevention (myelosuppression), and treatment expectations **Quality Audit Mechanism:** Real-time safety signal

tracking (particularly post-treatment HSCT outcomes)

14. Observation & Follow-Up Schedule

Total Duration: Extended post-treatment follow-up (tracking

survival up to 2-3 years) **Onsite Visit Cadence:** Days 1, 8, and 15

of every 28-day treatment cycle **Remote Monitoring Frequency:**

Monthly/Quarterly tracking during the post-treatment and long-

term survival phase **Checklist Review Interval:** Prior to every

cycle administration (reviewing liver function panels and
cytopenias)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				